

typical Sharpe ratio of 0 to 0.5 for a single asset using a trend-following strategy, which rises to 0.5 to 1.0 when looking at a portfolio of assets.

There are some potential problems, however, with the statistical properties of the Sharpe ratio. For example, rankings based on Sharpe ratios can be misleading if they are not adjusted for the impact of serial correlation. Researchers also usually compare differences in Sharpe ratios, which, due to the additive nature of variance error, is not as accurate as comparing Sharpe ratios of differences. Furthermore, the Sharpe ratio makes no distinction between upside and downside volatility. It penalizes equally both downside risk and upside return potential. Some researchers supplement their use of the Sharpe ratio with the Sortino ratio that uses only volatility below the mean. The Sortino ratio, however, discards all information on upside volatility and the right tail of the distribution, which may be useful to identify profit opportunities.⁹ For our internal work, we prefer to use a skewness-adjusted Sharpe ratio, which is easy to calculate and provides additional useful information.¹⁰ For our purposes here though, we will use the standard Sharpe ratio, since others are more familiar with it.¹¹

TAIL RISK AND MAXIMUM DRAWDOWN

For normal distributions, upside and downside volatility are approximately the same, but financial market returns are usually non-normally distributed. The difference between upside and downside volatility can be particularly problematic when returns are highly skewed, or nonsymmetric. Stock market returns are often negatively skewed, with an asymmetric left tail extending more toward negative values.¹² This creates tail risk, which can lead to greater-than-expected losses and aggravate the animal spirits mentioned earlier. Positive skewness is much preferred, since surprises should then work in our favor.

Academic research often just ignores tail risk. However, left tail risk, indicating negative skewness, is undesirable from a practitioner point of view. It can lead to large equity erosions, emotional distress, and untimely